

ALI YASSER KHUDHAYR

Electrical Engineering Student (Power & Machines) · KAU, Jeddah

Date of Birth: 10 January 2004 | **Mobile:** +966 57 033 6515
Nationality: Palestinian | **Email:** ali.y.khudhayr@gmail.com
Residence: Jeddah, Saudi Arabia | **Marital Status:** Single
Portfolio: ali-khudhayr.github.io

Personal Profile, Career Goal & Vision

Final-year Electrical Engineering student (Power & Machines) at King Abdulaziz University with a **4.68 / 5.00 GPA** and a Certificate of Academic Excellence. Experienced in power-system analysis, electrical machines, embedded systems and data analysis, with real work in team supervision, data reporting and teaching. Known for turning raw data into clear reports and for building practical digital tools. **Career goal:** to join a leading Saudi utility or energy company as a power systems engineer and earn professional registration with the Saudi Council of Engineers. **Vision:** to grow into a technical leader in Saudi Arabia's renewable-energy sector and contribute to the goals of Saudi Vision 2030.

Education

Bachelor of Science in Electrical Engineering — Power & Machines **Jan 2023 – May 2027 (Expected)**
King Abdulaziz University, Faculty of Engineering, Jeddah · GPA: 4.68 / 5.00 (Excellent)
Key courses: Power System Analysis, Power System Protection, High-Voltage Engineering, Electric Machines, Power Electronics, Automatic Control, Microcontrollers, Autonomous Systems.

Employment History

Operations & Data Reporting Assistant (Part-Time) **31 Jul 2026 – 28 Sep 2026**
Retail field-operations project, Jeddah, Saudi Arabia
- Prepared 14 purchase-invoice reports for retail branches, with cost per branch and per staff member.
- Built an Excel daily log for 9 staff at 10 sites in 5 cities with an automatic performance ranking.
- Delivered every report on a fixed weekly deadline, backed by photo evidence of each invoice.

Teaching Assistant **Aug 2025 – Dec 2025**
Department of Electrical and Computer Engineering, King Abdulaziz University
- Helped students master circuits, power-system concepts and simulation tools alongside course faculty.

Organizers Supervisor **Mar 2024 – Feb 2025**
Fitness Club, Deanship of Student Affairs, King Abdulaziz University
- Led 8 organizers across two daily shifts, managed attendance and coverage, and wrote weekly reports.

Industrial Field Visits **Aug 2025 – May 2026**
Al-Taqa Energy · Hala Company · Al-Rowad Factory · Savola Group (Afia)
- Observed power distribution, motor control centers, PLC automation and safety practices in live plants.

Projects Involved In

Smart Home Energy Management with Priority-Based Load Control **Aug 2026 – May 2027**
Senior Design Project (EE499), team of 3: appliance-level monitoring with automatic priority-based load shedding, monthly budget tracking and abnormal-load warnings.

Steering and Speed Calibration of the Quanser QCar 2 **Aug 2026 – Dec 2026**
Autonomous Systems (EE490), calibration lead: found that 2× throttle gives 2.17× speed and designed a 90-trial, 100 Hz Python experiment to cut route error by 30%.

Single-Phase SPWM Inverter — Hardware Prototype **May 2026 – Jun 2026**
Power Electronics (EE442): built a 12 V DC to 220 V, 60 Hz inverter using an Arduino UNO (200-step SPWM lookup), two MOSFETs and a step-up transformer; frequency error below 0.34%.

IoT Parking-Slot Occupancy Detection System **May 2026 – Jun 2026**
Microcontrollers (EE366): PIC18F4580 and ultrasonic sensor with three-state alerts and an ESP32 Wi-Fi link that sends time-stamped events to a web server; verified in Proteus.

Power Flow & Fault Analysis | Synchronous Motor V-Curves **Apr 2026 – May 2026**
EE451 & EE441: modeled a multi-bus network in PowerWorld and Simulink, and compared measured motor V-curves with a MATLAB/Simulink model.

Achievements

- Certificate of Academic Excellence ("Excellent"), Deanship of Student Affairs, KAU, 2024–2025.
- Designed a complete summer program for students, "Masarat Summer 2026".
- Competed in two Ideathon innovation challenges, placing third and then second.
- Maintained a 4.68 / 5.00 GPA while working part-time and volunteering every week.
- Completed more than 189 training hours and earned 50 certificates.

Voluntary Work

Active Member & Event Organizer, ECE Club, KAU — **Aug 2024 – May 2025**
Event Organizer, KAU Sports Hall (50 h) & KAU-ROBOCON 2023 — **May 2023 – Jan 2025**
200+ Volunteer Hours in education, events, community and charity work — **2019 – Present**
Ramadan Food Basket Distribution, Ith'am Food Bank — **May 2019 – Jun 2019**

Charity Work

Volunteer Teacher & Program Supervisor — **Jun 2024 – Present**
Qur'an Memorization Circle, Umm Al-Qura Mosque, Jeddah

- Teach and mentor students in Qur'an memorization, character building and Prophetic biography; plan weekly recitation schedules and track each student's progress.
- Created the circle's teaching tools: a reward web app, competitions in 7 categories and lesson presentations, also used in the "Masarat Summer 2026" program.

Certificates & Training

Introduction to Agentic Programming, Google Developer Group, KAU — **5 Aug 2026**
An Introduction to LaTeX Using Overleaf, Writing Center, KAU — **17 Feb 2026**
Image Generation Workshop, AI Club, KAU — **15 Feb 2026**
Qader: Cultural Program Design (12 h), Ajyal Association — **26 Jun 2025 – 28 Jun 2025**
Educator Competencies (15 h), Masarat Youth Association — **2 May 2024 – 6 Jun 2024**
Personal Effectiveness Series (4 courses), Human Capacity Center, KAU — **17 Jul 2023 – 2 Aug 2023**
FinTech Series: FinTech, Open Banking, Credit Cards, KAU — **25 Jan 2023 – 8 Feb 2023**
Advanced Self-Development Skills (60 h), Asas Center, TVTC — **10 Aug 2022 – 19 Aug 2022**
Microsoft Office 365: Word, Excel, PowerPoint (25 h), KAU — **25 Jul 2022 – 11 Aug 2022**
Al-Raed Volunteer Leader Program (48 h), KAU — **3 Jun 2022 – 27 Jun 2022**
Ideathon Creative Content Challenge (40 h), Ajyal Association — **3 May 2019 – 4 May 2019**

Professional Skills

- **Power & machines:** load flow, fault analysis, protection, machine testing, power electronics.
- **Simulation:** MATLAB, Simulink, PowerWorld, LTspice, Proteus, Quanser QLabs.
- **Programming & embedded:** Python (PyQt5), C/C++, Verilog, MATLAB; PIC18, Arduino, ESP32.
- **Office & tools:** advanced Excel reporting, Word, PowerPoint, LaTeX, AI and prompt engineering.
- **Engineering design:** WBS, Gantt charts, Pugh matrix, standards, technical reports.

Soft Skills

- Team leadership and supervision
- Teaching and public speaking
- Clear technical writing
- Problem solving under deadlines
- Time management and discipline
- Adaptability and fast learning

Hobbies

- Designing educational games and apps
- Creating motion-graphics videos
- Car mechanics and diagnostics
- Fitness training
- Reading history and biography
- 3D modelling for printing

Languages

Arabic: Native
English: Very good (language of study)

Other Activities

- AI-assisted software side projects
- Produced 29 explainer videos on 6 courses

References

Dr. Fahd Hariri
Senior Design Advisor
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Dr. Sultan Alghamdi
EE451 Instructor
smalgamdi1@kau.edu.sa

Dr. Omar Alnuqaity
IE201 Instructor
Industrial Eng., KAU

Website *Certificates*